

Adversarial-Robust GNNs for Cyber-Physical Anomaly Detection

Architecture and results — water, power and traffic networks

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The problem

Sensor networks on a graph, under attack

A water network is a **graph**: junctions carry a pressure time-series, pipes a flow time-series.

Sensing is **sparse and noisy**

- 50% of readings missing
- additive observation noise

An attacker **compromises a subset of sensors** and falsifies what they report.

The detector must, per snapshot:

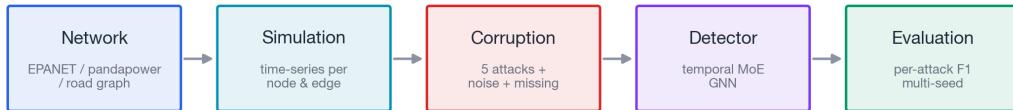
- **reconstruct** true pressure / flow
- **flag** which sensors are compromised

Attack	Knowledge	Goal
Random	black-box	disruption
Noise	black-box	disruption
Targeted	grey-box	local damage
Stealthy	white-box	evasion
Replay	grey-box	evasion

Five attack families spanning the standard knowledge / capability spectrum.

Pipeline

The same five stages for every domain



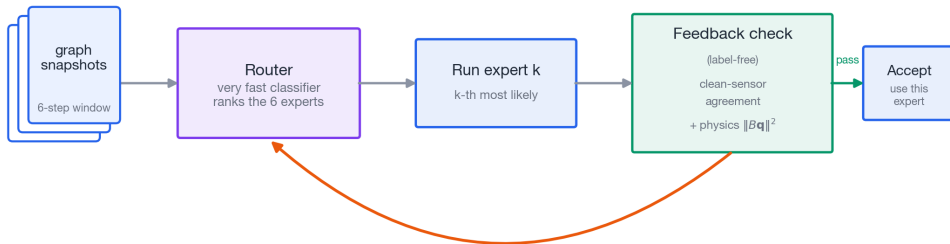
Only the first stage is domain-specific. Corruption, detector and evaluation are **shared, unmodified code**.

The architecture — cascade routing

An interpretable decision procedure, not an opaque blend

Cascade routing (router → expert → feedback → re-route)

interpretable: every decision shows which expert ran and why it was kept or rejected



negative feedback → next most likely expert

loop until an expert is accepted (or the ranking is exhausted)

The feedback mechanism

How the cascade decides, without labels

The router **rank**s the experts; the cascade runs one at a time and keeps or rejects it on a **label-free feedback check**:

1. run the router's most likely expert;
2. score it — do its “clean” sensors agree with their readings, and does the reconstruction obey mass conservation?
3. on negative feedback, re-route to the next-ranked expert (restricted to the router's **top-2**, for stability).

Every prediction carries **which** expert ran and **why** it was kept or rejected — an audit trail a soft blend cannot give.

Feedback signal (no labels)

$$s_k = \underbrace{\|\hat{p}_k - p\|_{\text{clean}}}_{\text{consistency}} + \underbrace{\|B\hat{\mathbf{q}}_k\|^2}_{\text{physics}}$$

standardised across experts. A confident, accurate router is trusted unless the feedback *clearly* disagrees.

The experts the cascade routes among

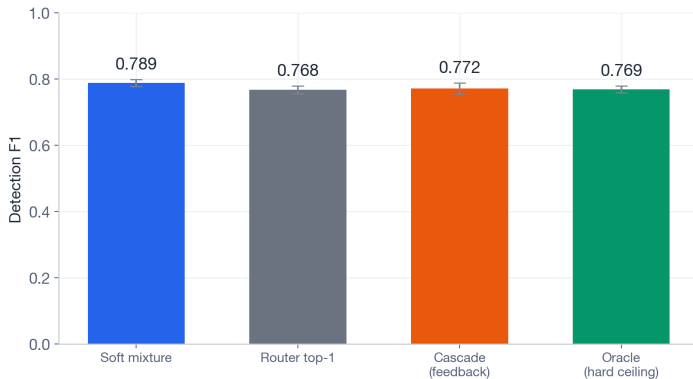
Six attack-specialised temporal GNNs (377 K parameters)

- Six experts, each a **GraphSAGE** → **GRU** model over the 6-step window, with four heads (reconstruct pressure/flow, flag anomalies).
- **Direct expert supervision** — every expert is trained on its own attack class regardless of routing, so a mis-routed class never starves for gradient.
- **Per-expert reconstruction** and a **physics loss** $\|B\mathbf{q}\|^2$ that forbids hydraulically impossible states.
- **Pattern features** — autocorrelation, adjacent-difference std, noise ratio — expose the observation noise a replayed value lacks.

The same six experts and the same physics term become the substrate for the power grid and the traffic network in Part 2 — unchanged.

Cascade vs the alternatives

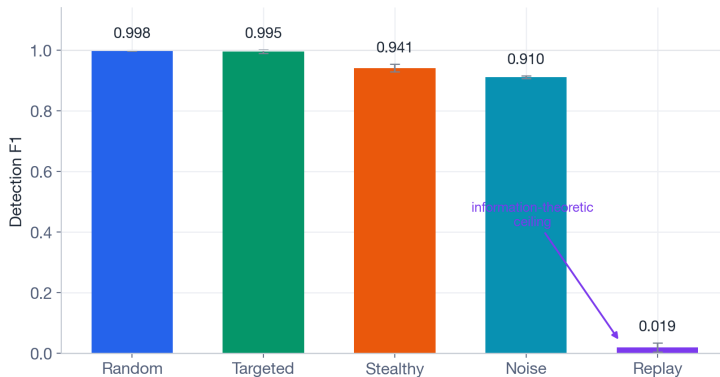
Modena, 5 seeds — the interpretable cascade matches the ceiling



The cascade reaches **0.772 ± 0.017** — matching the oracle ceiling (0.769, the best any hard routing could do) and within a point of a full soft blend (0.79). The experts are similar here, so the **interpretable cascade costs almost no accuracy**.

Detection results (Modena, 272 nodes)

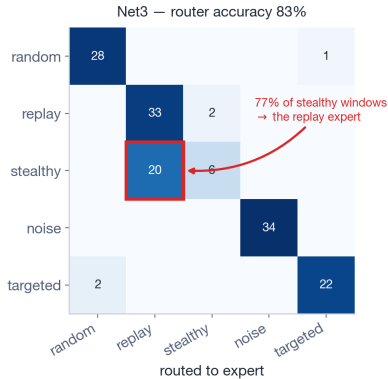
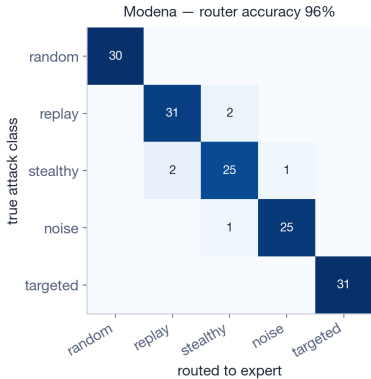
5 seeds, validation-calibrated threshold



Four of five attack families are detected reliably. **Replay sits near zero** — the next three slides are about why, how we recovered most of it, and why it is still not solved.

Diagnosing the router

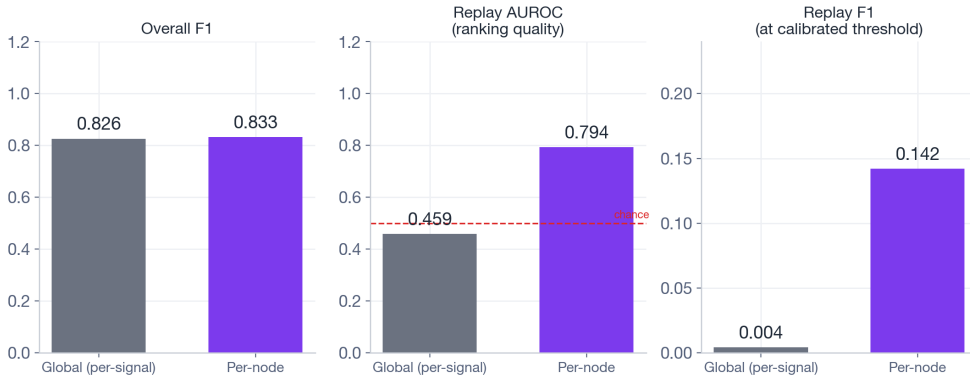
Where does a mis-routed attack go?



Modena's router is healthy (96%). On the larger **Net3** topology it sends **77% of stealthy windows to the replay expert** — both look smooth, and 97 nodes are too few to separate them. The cascade's feedback is designed to catch exactly this.

Recovering replay — normalisation

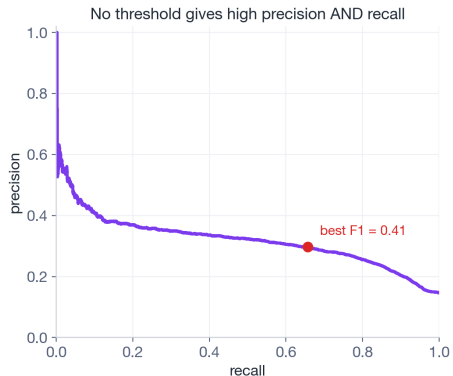
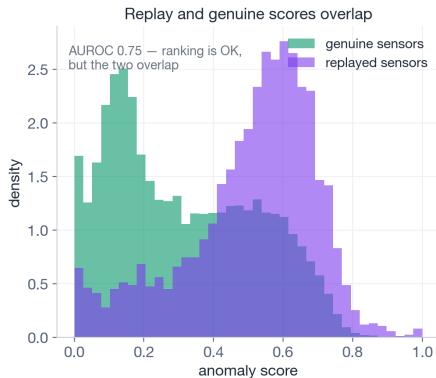
The supervisors' hypothesis, tested: water values sit too close together



Normalising each sensor by **its own** temporal scale amplifies the tiny per-node variation a replay lacks. Replay ranking jumps from **below chance to AUROC 0.79**; overall F1 holds (0.833).

Why replay still underperforms

Concrete evidence: good ranking, overlapping scores



Left: replayed sensors do score higher than genuine ones (AUROC 0.75) — but the distributions **overlap** heavily. **Right:** so no threshold gives high precision *and* recall; the best F1 is **0.41**. The residual signal is only the **missing observation noise** of a verbatim copy — a physical limit, not a training bug.

Part 2

Does any of this survive outside water?

One detector, three physics

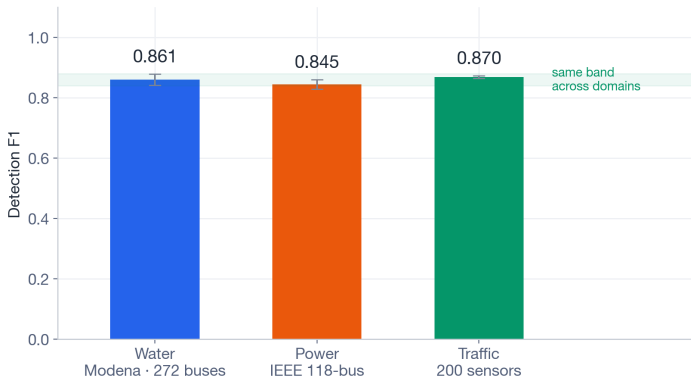
Model, training and corruption pipeline kept unchanged

	Water (Modena)	Power (IEEE-118)	Traffic (200 sensors)
Node / edge	junction / pipe	bus / line	detector / road
Node signal	pressure (m)	voltage (pu)	speed (km/h)
Edge signal	flow	active power	volume
Simulator	EPANET	pandapower power-flow	rush-hour + diffusion
Conservation	mass	Kirchhoff (KCL)	flow at junctions
Scale	272 / 317	118 / 186	200 / 675

Only a **per-domain data adapter** was written. The detector, the training script and the five-attack corruption pipeline are the **same code** in all three settings.

The detector is domain-agnostic

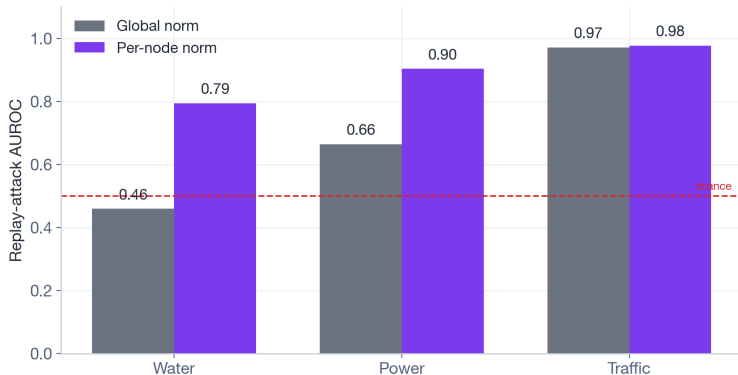
Overall F1, mean over seeds



Three unrelated physical systems, **no domain-specific tuning**, detection quality within a 2.5-point band.

The normalisation fix generalises

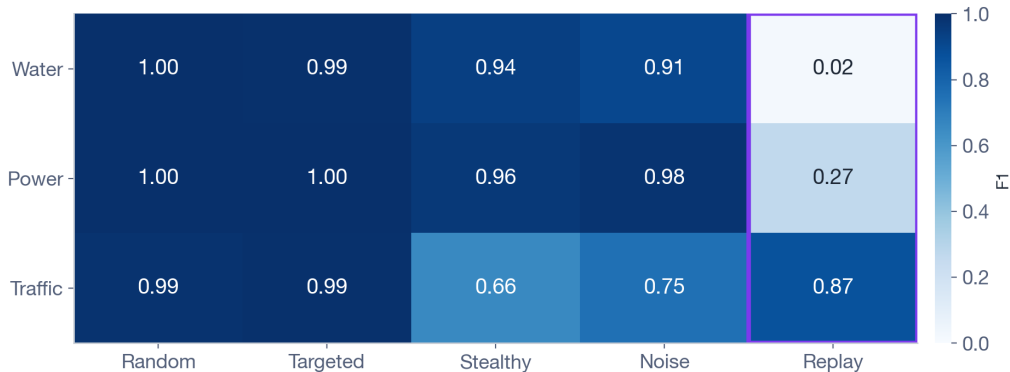
Grey bars = signal speed; purple = the fix



The grey bars already track **signal speed**: replay is hardest in slow-moving water (0.46) and trivial in fast-moving traffic (0.97). Per-node normalisation (purple) lifts it **most where it is hardest** — a principled, cross-domain fix, not a water-specific hack.

The difficulty profile inverts

Per-attack F1 across the three domains



Water and power find stealthy easy but replay hard; traffic is the **mirror image** — replay easy, stealthy blended into congestion. The domain, not the model, decides the hard case.

Summary

Part 1 — Water

- **Cascade routing**: interpretable, matches the ceiling (0.772), stable
- Router failure mode diagnosed and contained
- **Per-node normalisation** recovers replay ranking (AUROC 0.46 $\rightarrow$ 0.79)
- Replay's residual gap explained: overlapping scores, a physical limit

Part 2 — Power & traffic

- Identical detector transfers unchanged
- **F1 0.82 – 0.89** across three physical systems
- The normalisation fix **generalises**: replay AUROC 0.66 $\rightarrow$ 0.90 on power
- Replay difficulty tracks signal speed — a general principle

Next steps

- (1) Push the cascade onto networks where the experts differ more, so re-routing also *gains* accuracy.
- (2) Extend the threat model toward the smart-grid FDIA and transport-security literature.
- (3) Consolidate the three domains into one evaluation and identify a target venue.

Thank you

Questions?